

EAJRS Cross Talk Series – Kick-off Event Report

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Beyond Transliteration: Romanisation and Word Segmentation in Japanese Metadata

Kick-off Event Report

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This report has been prepared based on meeting notes transcribed by Notta.ai.

Panellists

- Toshinobu Ogiso (National Institute for Japanese Language and Linguistics)
- Tomokazu Takada (National Institute for Japanese Language and Linguistics)
- Katsunori Iino (National Institute of Informatics)
- Nobutake Kamiya (University of Zurich Library; EAJRS Board Member)
- Chris Dillon (British Library; EAJRS Board Member)

Chair: Yasuyo Ohtsuka (British Library)

About the Round Table

This round table was held as the kick-off event of the EAJRS Cross Talk Series, under the theme **‘Why Romanisation and Word Segmentation in Japanese, and Why Now?’**

As bibliographic information and metadata relating to Japanese materials are increasingly shared, searched and used internationally, there is a growing need to reconsider the role of Japanese romanisation. Even as AI-powered translation continues to advance, there remains a need for systems that can connect information that is difficult to translate—such as proper names and authority data—with the original language and make it discoverable and shareable internationally.

The discussion therefore approached romanisation not simply as a means of converting Japanese into the Latin alphabet, but through two fundamental questions: **who is using romanisation and for what purpose, and where should Japanese words be segmented?**

Part 1: Who Is Romanisation For, and What Is It For?

Takada argued that the first step in considering romanisation is to clarify **who will use it and for what purpose**. Romanisation serves different functions when used by native Japanese speakers, when providing pronunciation guidance to people who cannot read Japanese, and when supporting searching, identification and data linking in library metadata.

He also noted that the users of romanised data are not limited to people. Search systems, databases and AI systems are also users of such data. Consequently, the preferred approach may differ depending on whether the priority is readability, pronunciation, discoverability or machine processing.

Iino emphasised the need for romanisation that is accessible to non-Japanese speakers in an international information environment. While international standards such as the ALA-LC romanisation system have clear value, he noted that no single system can necessarily be regarded as the only solution for all users and linguistic environments.

Ogiso pointed out that the new Japanese Cabinet notification on romanisation is primarily concerned with the representation of proper names and does not provide comprehensive guidance on the romanisation of longer Japanese texts or on word boundaries. This means that applying romanisation to general Japanese metadata involves issues that cannot be resolved by a transliteration system alone.

Kamiya explained that, for libraries outside Japan, providing romanised metadata for Japanese materials is often essential. If individual institutions adopt different systems, discoverability can suffer, making a degree of commonality desirable. At the same time, he described practical situations in which multiple standards are combined — for example, using the DIN [Deutsches Institut für Normung] standard for romanisation while referring to NII guidelines for word segmentation in German-speaking library environments.

Dillon also discussed the role of romanisation in Japanese-language learning. Since kanji acquisition can represent a significant burden at the early stages of learning Japanese, the use of romanisation can allow learners to focus initially on grammar and basic sentence patterns, potentially supporting continued engagement with the language.

The differences between the Hepburn and Kunreisiki systems were also discussed. Hepburn romanisation has traditionally placed greater emphasis on making pronunciation accessible to English speakers, whereas Kunreisiki is more closely

aligned with the systematic representation of the Japanese phonological system. It was noted, however, that the intuitive nature of Hepburn romanisation is partly rooted in an English-language perspective and may not be equally intuitive for speakers of languages such as French, Chinese or Korean.

The discussion thus highlighted the diversity of purposes and users of romanisation, and the shared understanding that **a single ‘correct’ system cannot by itself address all requirements.**

Part 2: Romanisation and Word Segmentation

The second part of the discussion focused on the close relationship between romanisation and word segmentation.

Ohtsuka used place names such as *Yurakucho* and *Shinjuku-ku* as examples of cases in which decisions must be made about whether elements should be written together or separated, and whether a hyphen should be used. She suggested that the more fundamental issue is determining **where word boundaries should be placed in Japanese.**

Using the Preamble to the Constitution of Japan as an example, Ogiso demonstrated that in Japanese text written with a mixture of kanji and kana, kanji provide important visual clues to word boundaries. When the same text is written entirely in kana or in romanisation, however, these clues are greatly reduced, making segmentation more difficult. In speech, intonation and pauses provide additional information, but such information is not directly represented in writing. Word segmentation therefore becomes particularly important in phonetic writing systems.

Historical and educational examples were also introduced, including the 16th-century *Amakusa-bon Heike Monogatari* and textbooks for younger primary-school pupils. These examples demonstrate that when Japanese is represented using phonetic scripts, some form of word segmentation is essential for readability.

Romanisation should therefore not be understood simply as a process of character conversion. It also involves **linguistic analysis to determine where Japanese words and units should be segmented.**

Challenges in Applying Rules and Using AI

The practical difficulties involved in establishing and applying segmentation rules were also discussed.

The ALA-LC system contains relatively detailed rules for Japanese word segmentation, extending to approximately 35 pages. Ogiso explained that substantially larger bodies of linguistic rules may be required when attempting to segment Japanese systematically at the word level for purposes such as corpus construction. Even highly detailed rules can leave room for ambiguity, particularly when multiple rules overlap or exceptions need to be addressed.

Iino noted that the NII cataloguing rules also contain areas that are undefined or potentially inconsistent. In practice, cataloguers may therefore rely on previous examples and their own experience when making decisions. Such reliance on individual expertise presents a challenge for further automation.

Examples were also discussed in which generative AI has been used to produce Japanese romanisation with a high degree of accuracy. However, where detailed decisions concerning capitalisation, the representation of numbers or the treatment of proper names have not been explicitly codified, consistent automated processing remains difficult.

Takada emphasised that romanisation should not be regarded simply as a process of replacing characters. It involves analysing the structure of Japanese and determining word boundaries. He therefore highlighted the potential value of morphological analysis and AI-based tools in reducing the workload for practitioners.

Beyond a Single ‘Correct’ System: Building Bridges between Systems

Throughout the discussion, there was a shared understanding that the immediate objective is not to determine which romanisation system is superior, nor to impose a single set of rules.

Rather, an important goal is to establish **‘bridges’ or mappings between different romanisation systems and metadata representations.**

Librarians, researchers, teachers and learners may use different systems for legitimate reasons. Instead of attempting to eliminate this diversity, the challenge is to ensure that users can reach the same resources regardless of which system they use to search.

Developing mappings between different forms of romanisation and metadata would therefore make it possible to retain diversity while improving international discoverability and data interoperability.

From this perspective, Japanese romanisation should not be regarded solely as a transliteration issue. It is also a question of **metadata design and infrastructure for connecting Japanese materials to the international information environment**.

Future Directions

The discussion confirmed the importance of continuing to share the practical challenges faced by professionals working with Japanese metadata in libraries, museums and other cultural heritage institutions, and of continuing to examine the future of Japanese romanisation and word segmentation.

Future discussions will need to address specific practical questions, including how particles, auxiliary verbs and affixes should be treated; how capitalisation should be handled; and when hyphens should be used in addresses and proper names.

The panellists also highlighted the importance of avoiding an ever-expanding body of rules. Instead, tools such as AI and morphological analysis (e.g. Web ChaMame) could be used to support practitioners, reducing the burden of manual decision-making while maintaining an appropriate degree of consistency.

The next Cross Talk will examine practical examples of transliteration and word segmentation from organisations including NII, NDL and OCLC, with the aim of developing the discussion further through concrete examples of current practice.

Key Points

- **Romanisation serves multiple purposes.**
Its role differs depending on whether it is being used for Japanese-language learning, pronunciation guidance, library discovery and identification, metadata interoperability or processing by AI and databases.
- **The central issue is where to place word boundaries.**
In Japanese, kanji provide important visual clues to word boundaries. When Japanese is written in romanisation or kana alone, these clues are reduced, making word segmentation essential.
- **Romanisation systems vary across countries, regions and professional contexts.**
Hepburn, Kunreisiki, ALA-LC and DIN each have different backgrounds and purposes. Rather than seeking to impose a single system, it may be more productive to develop mechanisms that connect different systems.

- **Mappings between systems are key.**

By linking different forms of romanisation and metadata, users should be able to discover the same resources regardless of the system they use when searching.

- **AI and morphological analysis offer significant potential.**

Romanisation is not simply character conversion; it involves analysing Japanese structure and determining word boundaries. AI-based tools could help reduce the burden on practitioners while supporting consistency.

Conclusion

Thinking about Japanese romanisation means more than simply converting Japanese into the Latin alphabet. It requires consideration of **who will use the data, what they will use it for, and where Japanese should be segmented.**

The discussion highlighted the importance of moving beyond the search for a single romanisation system. Instead, the focus should be on accommodating different systems, connecting them through mappings and making effective use of AI and other technologies to create an environment in which Japanese materials can be discovered, shared and linked internationally.